

Phase 1: overall comments

Section 6, Group G (customer) - wrote for developer Group H | Electronics Retail (warranty)

Plenty here to build on: six clearly identified services, a warranty pain point tied to a physical artefact, and business-model figures with real teeth - concrete margins, specific repair fees in NIS. Staying with the current manual reality throughout (paper logs, phone calls, carbon invoices) rather than drifting into future-system territory is exactly the right discipline. The capacity table is a different story - abbreviations nobody has defined, a compound figure that can't be decomposed, and a workforce total that doesn't add up to the number written. Processes also stop at the happy path every time; at minimum you need exception steps for failed repairs, rejected claims and unavailable stock.

A note to your developer group (Group H)

Before building on this, confirm with the client exactly what 'Active orders' counts (repair jobs? sales orders? both?), and get the per-branch inventory and repair figures as separate named metrics rather than the compressed shorthands currently in the table. You also need to clarify the exception paths for warranty claims and hardware repairs - the happy paths are workable, but you cannot design any meaningful workflow without knowing what happens when things go wrong.



Faculty of Engineering and Technology | Computer Science Department
COMP433 Software Engineering | Section 6 | Semester 2 (Spring) 2025/2026

Phase 1 — Task 1.1

Electronics Retail with warranty tracking — Business Description

Prepared by

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Group No: G | Role: Customer Group

Developer Group: H | Business No: 61

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1. Business overview

The business is a retail electronics store with branches in Ramallah and Birzeit, serving students and residents. It sells hardware (phones, computers, accessories) and provides manual maintenance and warranty services. Operations currently rely on physical logs, paper invoices, and direct phone communication between staff and branches for inventory and repair tracking.

Two named locations pins the scope immediately - exactly the kind of grounding detail the brief asks for.

Describing what actually exists now, rather than imagining what a system might do, is the right instinct - it keeps the business definition honest.

2. Business services

Service 1 - Counter sales & product consultation: Staffed counters provide in-person sales and consultation for electronics. Associates perform live product demonstrations and suggest specific models based on customer requirements and budget.

Nice elicitation here - live demos, budget-based recommendations... this is the operational texture that makes a description really useful.

Service 2 - Hardware repair & technical servicing: Technicians perform on-site diagnostics and repairs for hardware and software issues. Customers drop off devices and must call the branch to check repair status against manual staff logs.

Service 3 - Warranty registration & claim management: Associates manually record product serial numbers on physical purchase agreements and invoices. Customers must present these paper invoices at the service desk to verify warranty coverage or file manual claims.

Tying a pain point to a physical artefact is exactly what motivates a build. That connection is doing real work.

Service 4 - Trade-in program: Customers trade in used electronics for store credit or discounts. Staff manually assess the device's physical condition on-site to provide an immediate valuation toward a new purchase.

Service 5 - Returns & exchange processing: Staff process product returns by manually verifying paper records and warranty status. Associates physically inspect items for eligibility before authorizing manual refunds or exchanges.

Service 6 - Multi-branch inventory check & reservation: Staff check stock at other branches by making manual phone calls to confirm physical shelf availability. If found, items are reserved via verbal confirmation and transferred between branches by a delivery driver.

A delivery driver moves stock between branches, yet no driver appears in the employee table. That gap needs fixing before anyone can trust the staffing picture.

3. Capacity and statistics

Table 3.1: General Business Metrics

Metric	Value	Branch	Ramallah	Birzeit
Monthly Customers	2800	Floor Space	140 sqm	90 sqm
Active orders	20 Requests	Daily Traffic	71	29
Catalog items	750(720 HW/30 services)	Inv/Repair	850Units/90	320U/35C

'20 Requests' - requests of what? This cell is undefined; a developer cannot size throughput from a bare number with no label.

The '850Units/90' entry is unreadable as a compound - separate and label the inventory figure and the repair figure, or they are unusable to anyone trying to plan capacity.

Products: Phones (250), Laptops (180), Wearables (85), Gaming (80), Network (70), Acc. (55), Services (30)

4. Business processes

1. Counter Sales & Product Consultation: 1. Customer explains needs and budget. 2. Associate recommends and compares products. 3. Demo provided if needed. 4. Customer selects product. 5. Stock confirmed. 6. Payment made, invoice issued, item handed over.

'U' and 'C' are never expanded anywhere in the document. Stock and repair counts for Birzeit cannot be reconciled from these abbreviations.

2. Hardware Repair & Technical Servicing: 1. Customer brings device and explains issue. 2. Staff records details and diagnoses fault. 3. Cost and time explained. 4. Customer approves repair. 5. Device repaired and tested. 6. Customer calls for updates. 7. Device collected and charges paid.

This does sum correctly to 750, which is a welcome consistency compared to the rest of the table - so you clearly can do it.

3. Warranty Registration & Claim Management: Registration: 1. Customer buys product. 2. Agreement prepared. 3. Customer signs. 4. IMEI/serial recorded. 5. Warranty registered, invoice issued. **Claims:** 1. Customer presents invoice. 2. Warranty checked. 3. Issue reviewed. 4. Product accepted if approved. 5. Customer returns later.

What happens if the repair cannot be completed, or the device is beyond economic repair? That branch is entirely absent, and it is not a rare edge case.

No rejection branch, and 'returns later' tells a developer essentially nothing. The unhappy path needs spelling out - even a sentence per exception is enough.

4. Trade-in Program: 1. Customer brings used device. 2. Staff inspects item. 3. Trade-in value offered. 4. Value applied as credit/discount. 5. Customer buys new product.

5. Returns & Exchange Processing: 1. Customer brings item and receipt. 2. Policy and records checked. 3. Item inspected. 4. Refund, exchange, or credit approved. 5. Transaction completed. 6. Item restocked or returned.

6. Multi-Branch Inventory Check & Reservation: 1. Customer requests unavailable item. 2. Product details collected. 3. Other branches contacted. 4. Stock confirmed. 5. Item reserved. 6. Transfer arranged if needed. 7. Customer notified for pickup.

5. Employees and roles

	Ramallah	Birzeit	Main Responsibilities
General Manager	*1	-	Price catalogs, global supply, and financial oversight.
Branch Manager	1	1	Daily staff management, manual cash reconciliation.
Sales Associate	3	2	Product consultation and writing carbon paper invoices.
Warranty Coordinator	1	-	Managing physical filing binders and IMEI verification.
Maintenance Tech	2	1	Hardware repairs and updating manual "Job Sheets".
Cashier	1	1	Payment processing, "PAID" stamps, and cash drawer.

The 'carbon paper invoices' detail is vivid & specific - but it blurs the service mapping slightly. Is writing invoices really the Associate's defining responsibility in this context, or is it incidental to the actual service?

Cashier is a payment-handling role, not a service. Listing it alongside the six business services mixes levels of description - these things belong in different parts of the document.

Total workforce across all branches: 13 employees.

6. Business model

Main income:

- **Product sales margin:** 10% to 30% markup on hardware; higher margins on accessories.
- **Warranty protection plans:** Sale of physical 12-month extended warranty at checkout.

Ramallah column sums to 9, Birzeit to 5 - that is 14, not 13. Sort this out before sizing anything, and account for the delivery driver who is mentioned elsewhere.

A clearly scoped income stream with a defined product - this is the specificity that elicitation should produce, and it does the job.

Secondary income:

- **Repair service fees:** Flat labor charges (e.g., 50 NIS for screens, 35 NIS for batteries) plus spare part markups.
- **Trade-in price spread:** Profit from the difference between used device acquisition and refurbished resale price.
- **Inter-branch transfer fees:** 20 NIS flat fee for manual device delivery between branches.
- **Marketing & shelf placement:** Fixed fees for featuring brands on display tables or in brochures.

Exact figures rather than vague ranges - precisely the kind of detail that makes a business model credible & usable by anyone costing up a build.

7. Challenges and problems

- **Manual Tracking:** Paper invoices lead to lost records and incorrect warranty data.
- **Communication Gaps:** Inventory and repair updates require phone calls, causing delays.
- **Processing Slowness:** Manual verification of history stalls returns and exchanges.
- **Human Error:** High volume and manual logging cause mistakes in stock and sales records.

Phase 2 feedback

Section 6, Group G - Used Electronics (developer role)

Plenty here to build on: the account-lockout rule (five attempts, fifteen-minute window) is textbook-precise, the cost section has a confidence band and an honest exclusions list, and several SR refinements name specific fields with exact error messages. The problem is that the URs above them have not been held to the same standard - UR8, UR9 and UR12 bury multiple distinct user needs behind 'support ... management', and UR12/SR12 point at an unstated external policy instead of spelling out observable refund conditions. UR22 and its SRs also need 'shall' throughout. Carry that SR-level precision upward into Phase 3 use-case modelling.

Phase 2 evaluation

User Requirements: B System Requirements: B Effort & Cost: B Overall: B
SR precision & cost reasoning impress; URs need decomposing and modality tightening.



Faculty of Engineering and Technology

Computer Science Department

Software Engineering - COMP433

Semester 2 (Spring) 2025/2026

Phase 2: Requirements Engineering

Used-Electronics Exchange Platform

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Business No: 62

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Business introduction

The Used Electronics Exchange Platform will cater to the needs of a business that offers buying, selling, exchanging, repairing, inspecting, and delivering of second-hand electronic devices via two branches in Ramallah and Nablus. The platform is intended to help streamline the processes, minimize manual documentation, and manage the devices, transactions, and other services.

Some of the key stakeholders of the platform are clients, employees who sell devices, those who inspect devices, technicians, employees responsible for deliveries, managers, accountants, and others. Users of the platform need to perform certain actions in connection with device trading, maintenance, inspections, delivery services, and other aspects related to the business management.

The core services that will be provided by the business are buying and selling of used devices, trading of devices, making appointments with users, inspecting and assessing devices' quality, repairing devices, managing delivery services, handling client complaints, and generating various reports from the business records. The assumptions made for creating these requirements are that all devices have to be inspected prior to the selling or exchange, user authentication is needed to access the protected resources, and that business records need to be managed at multiple locations.

Below are the required requirements of the Used Electronics Exchange Platform.

1. User Requirements (UR list)

Device Buying, Selling, and Exchange

- UR1. The system shall allow customers to browse and search available devices.
- UR2. The system shall allow customers to purchase available devices.
- UR3. The system shall allow customers to submit devices for sale or exchange.
- UR4. The system shall allow employees to record and manage device information before devices are offered for sale or exchange.

Device Inspection

- UR5. The system shall support device inspection and condition assessment activities.

'Support ... activities' tells us the theme but not what the system actually does. An inspector reading this cannot picture a single screen or action. Rewrite it so the UR describes what the user needs to accomplish, not the domain area it sits in.
- UR6. The system shall provide customers with inspection information for inspected devices.

Appointment Management

- UR7. The system shall allow customers to manage appointments for inspections and exchanges.

Your own margin note already sees the problem: 'manage' is hiding at least three distinct user needs. SR7.1-7.3 already split them into create, modify, cancel - UR7 should do the same.

Repair Services

- UR8. The system shall support repair request management.

'Support repair request management' is a section title dressed up as a requirement. What can a user do - submit, update, track, close? Without any observable behaviour in the UR itself there is nothing to accept or reject.

Delivery Services

- UR9. The system shall support delivery request management.

Same issue as UR8 - 'management' silently bundles submit, view, update and notify. You wouldn't write a single acceptance test for all four of those, so why write a single UR?

Customer Account and Access

- UR10. The system shall allow users to authenticate before accessing protected services.

Payments and Refunds

UR11. The system shall support payment processing for purchases, exchanges, and repair services.

UR12. The system shall support refunds according to the business refund policy.

Pointing at an unquoted external policy leaves the tester with nothing to check. What conditions must hold for a refund to be approved? The UR needs to state them, or attach the policy as a numbered appendix.

Complaints Management

UR13. The system shall allow customers to submit complaints and track their resolution status.

Records and Reporting

UR14. The system shall maintain customer transaction records and device ownership records.

UR15. The system shall provide business performance reports to authorized managers.

Domain Requirements

UR16. The system shall require device inspection before a device can be offered for sale or exchange.

UR17. The system shall classify inspected devices according to predefined condition categories.

UR18. The system shall maintain warranty information for devices when applicable.

'When applicable' is doing a lot of quiet work here. Which categories carry warranties? Without scoping that, a developer cannot tell when to enforce warranty storage and when to skip it.

Non-Functional Requirements

UR19. The system shall maintain secure access to business data according to user roles and permissions.

UR20. The system shall retain historical business records.

Retain them for how long? There is no retention period here. A system that deletes everything after a week technically satisfies this as written - you need a minimum duration to make it testable.

UR21. The system shall support simultaneous appointment booking requests from multiple users without creating duplicate appointment records.

UR22. The system should minimize duplicate data entry and unnecessary navigation during common customer and employee tasks.

'Minimize' points in a direction but does not give you a finish line. How few steps, how few repeated fields, would actually pass? Without a threshold you cannot demonstrate compliance, and a developer cannot know when to stop.

UR22 uses 'should' for what is clearly a mandatory usability obligation. Swap it to 'shall' - otherwise a supplier can leave the navigation inconsistent and argue they technically met the spec.

2. System Requirements (SR Refinements)

UR1 Refinements

SR1.1 The system shall allow customers to search for devices using keywords.

SR1.2 The system shall allow customers to filter devices by category, brand, condition category, branch location, and price range.

SR1.3 The system shall display the device name, brand, model, condition category, selling price, warranty status, and availability status for each listed device.

SR1.4 The system shall display a notification when no devices satisfy the selected search criteria.

SR1.5 If the price range minimum exceeds the maximum, the system shall display a field-level validation error: "Minimum price must be less than or equal to maximum price" and not submit the search.

UR2 Refinements

SR2.1 The system shall allow authenticated customers to purchase devices whose status is Available.

SR2.2 The system shall verify device availability before initiating payment processing.

SR2.3 The system shall create a transaction record containing a transaction identifier, customer identifier, device identifier, payment amount, and transaction date after successful payment.

SR2.4 The system shall update the device status to Sold immediately after successful payment confirmation.

SR2.5 The system shall cancel the transaction and notify the customer if payment authorization fails.

SR2.6 The system shall generate a purchase receipt containing the transaction identifier, purchased device information, payment amount, and purchase date.

UR3 Refinements

SR3.1 The system shall allow authenticated customers to submit devices for sale.

SR3.2 The system shall allow authenticated customers to submit devices for exchange.

SR3.3 The system shall require device category, brand, model, estimated age, condition description, and at least one image before accepting a submission.

SR3.4 The system shall assign a unique submission identifier to each submitted device.

SR3.5 The system shall assign the status Pending Inspection to every newly submitted device.

UR4 Refinements

SR4.1 The system shall allow authorized employees to create device records.

SR4.2 The system shall allow authorized employees to update device records.

SR4.3 Each device record shall contain a unique device identifier.

SR4.4 The system shall record the employee identifier and modification date for each device record update.

UR5 Refinements

SR5.1 The system shall allow authorized inspectors to record inspection results for submitted devices.

SR5.2 Inspection records shall contain the inspection date, inspector identifier, condition category, identified defects, and inspection outcome.

SR5.3 The system shall prevent devices that have not passed inspection from being offered for sale or exchange.

SR5.4 The system shall record the reason for inspection failure when a device does not pass inspection.

UR6 Refinements

SR6.1 The system shall allow customers to view inspection information for inspected devices.

SR6.2 Inspection information shall include inspection date, condition category, inspection outcome, and identified defects.

SR6.3 The system shall prevent customers from modifying inspection information.

UR7 Refinements

SR7.1 The system shall allow authenticated customers to create appointments.

SR7.2 The system shall allow customers to modify future appointments.

SR7.3 The system shall allow customers to cancel future appointments.

SR7.4 Appointment records shall contain appointment identifier, appointment type, date, time, branch location, and customer identifier.

SR7.5 The system shall reject any appointment request whose selected date, time, and branch are already assigned to another confirmed appointment.

UR8 Refinements

SR8.1 The system shall allow customers to submit repair requests.

SR8.2 Repair requests shall contain device information, problem description, submission date, and customer identifier.

SR8.3 The system shall assign a unique repair request identifier to each repair request.

SR8.4 The system shall allow authorized technicians to update repair requests with one of the following statuses: Pending, In Progress, Completed, or Rejected.

SR8.5 The system shall allow customers to view the current status of their repair requests.

UR9 Refinements

- SR9.1** The system shall allow customers to request delivery during a purchase transaction.
 - SR9.2** Delivery records shall contain delivery identifier, delivery address, contact number, delivery date, and delivery status.
 - SR9.3** The system shall allow authorized delivery employees to update delivery status.
 - SR9.4** The system shall notify customers whenever the delivery status changes.
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UR10 Refinements

- SR10.1** The system shall require users to authenticate using a registered email address and password before accessing protected services.
 - SR10.2** The system shall deny access to protected services if the submitted credentials do not match a registered user account.
 - SR10.3** The system shall store passwords using a secure one-way hashing algorithm.
 - SR10.4** The system shall lock a user account after five consecutive failed login attempts within a fifteen-minute period.
 - SR10.5** The system shall allow users to reset their passwords through a verified email address.
-

SR10.3 names a mechanism, not an obligation. The requirement is that passwords must not be recoverable or disclosed - how that is achieved is a design decision for the architect, not something to mandate in the spec.

This is exactly the right level: a named threshold (five attempts), a recovery window (fifteen minutes), directly testable. More of this precision earlier in the security section would have lifted the whole chapter.

UR11 Refinements

- SR11.1** The system shall process payments for purchases, exchanges, and repair services.
- SR11.2** Payment records shall contain payment identifier, payment amount, payment method, payment date, and transaction identifier.
- SR11.3** The system shall verify payment completion before confirming any transaction.
- SR11.4** The system shall generate an electronic receipt after successful payment.
- SR11.5** The system shall record failed payment attempts for auditing purposes.
- SR11.6** The system shall not confirm a purchase, exchange, or repair transaction until payment has been successfully completed.

UR12 Refinements

SR12.1 The system shall allow authorized employees to process refunds according to the business refund policy.

SR12.2 Refund records shall contain refund identifier, refund amount, refund reason, approval employee identifier, and refund date.

SR12.3 The system shall reject refund requests that do not satisfy the business refund policy.

This SR inherits the deferral from UR12 rather than resolving it. 'Does not satisfy the policy' is not a test condition - what failure states should trigger rejection? List them here.

SR12.4 The system shall update the corresponding transaction record after a refund is completed.

UR13 Refinements

SR13.1 The system shall allow customers to submit complaints electronically.

SR13.2 Complaint records shall contain complaint identifier, complaint description, submission date, customer identifier, and complaint status.

SR13.3 The system shall allow authorized employees to assign one of the following statuses to a complaint: Open, Under Review, Resolved, or Closed.

SR13.4 The system shall allow customers to view the current status of submitted complaints.

SR13.5 The system shall maintain a history of complaint status changes.

UR14 Refinements

SR14.1 The system shall maintain transaction records for purchases, sales, exchanges, repairs, and refunds.

SR14.2 The system shall maintain ownership history records for each device.

SR14.3 The system shall allow authorized employees to retrieve records using customer identifier, device identifier, transaction identifier, or date range.

SR14.4 The system shall restrict record deletion privileges to managerial roles.

UR15 Refinements

SR15.1 The system shall allow authorized managers to generate sales reports.

SR15.2 The system shall allow authorized managers to generate inspection reports.

SR15.3 The system shall allow authorized managers to generate repair reports.

SR15.4 The system shall allow authorized managers to generate delivery reports.

SR15.5 Reports shall support filtering by branch location and date range.

SR15.6 Reports shall be exportable in PDF format.

PDF is a fair concrete constraint - it is at least testable. Worth confirming whether the customer asked for this specifically, or whether it is your own design preference that has drifted into the requirements.

UR16 Refinements

SR16.1 The system shall prevent a device from being listed for sale before inspection completion.

SR16.2 The system shall prevent a device from being listed for exchange before inspection completion.

UR17 Refinements

SR17.1 The system shall classify inspected devices using predefined condition categories.

SR17.2 Only authorized inspectors shall assign or modify device condition categories.

SR17.3 The system shall store the assigned condition category as part of the inspection record.

UR18 Refinements

SR18.1 The system shall store warranty start date and warranty expiry date for devices covered by a warranty.

SR18.2 The system shall allow customers to view warranty information for devices offered for sale.

SR18.3 The system shall prevent entry of a warranty expiry date earlier than the warranty start date.

UR19 Refinements

- SR19.1** The system shall enforce role-based access control for protected resources.
 - SR19.2** User roles shall include Customer, Sales Employee, Inspector, Technician, Delivery Employee, Accountant, and Manager.
 - SR19.3** The system shall prevent users from accessing functions assigned to roles other than their own.
 - SR19.4** The system shall record unauthorized access attempts in an audit log.
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UR20 Refinements

- SR20.1** The system shall retain transaction, inspection, repair, delivery, and complaint records after completion.
 - SR20.2** The system shall allow authorized users to retrieve historical records.
 - SR20.3** The system shall prevent unauthorized deletion of historical records.
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UR21 Refinements

- SR21.1** The system shall maintain correct appointment records when multiple customers attempt to book appointments simultaneously.
 - SR21.2** The system shall ensure that no appointment slot is assigned to more than one customer.
 - SR21.3** The system shall complete appointment creation as a single operation to prevent duplicate bookings.
 - SR21.4** The system shall reject a booking request and notify the customer if a scheduling conflict is detected.
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'Single operation' hints at a transactional implementation rather than specifying what must actually hold. Say the thing itself: no appointment slot may be assigned to more than one customer simultaneously, regardless of concurrent requests.

UR22 Refinements

SR22.1 The system should provide a navigation structure that is consistent across all customer and employee functions.

SR22.1-22.3 all inherit 'should' straight from UR22, so the modality error has propagated downward. If consistent navigation is a real requirement, every one of these needs 'shall'.

SR22.2 The system should display validation messages when users submit incomplete or invalid information.

SR22.3 The system should minimize duplicate data entry during common customer and employee tasks.

3. Effort and Cost Estimation

Per-UR Estimate

UR	Description (short)	Concurrent Developers	Person-Weeks per Developer	Total Effort (Person-Weeks)
UR1	Browse and search available devices	2	2.0	4.0
UR2	Purchase devices and complete payment process	2	3.0	6.0
UR3	Submit devices for sale or exchange	2	2.5	5.0
UR4	Record and manage device information	1	2.0	2.0
UR5	Device inspection and condition assessment	2	2.5	5.0
UR6	View inspection information	1	1.0	1.0
UR7	Manage appointments	2	3.0	6.0
UR8	Manage repair requests and status tracking	2	3.0	6.0
UR9	Manage delivery requests and status tracking	2	2.5	5.0
UR10	User authentication	2	2.0	4.0
UR11	Payment processing	2	3.0	6.0
UR12	Refund processing	1	2.0	2.0
UR13	Complaint management	1	1.5	1.5

UR14	Transaction, ownership, and warranty records	1	2.0	2.0
UR15	Business performance reports	2	2.0	4.0
UR16	Mandatory inspection before sale or exchange	1	1.5	1.5
UR17	Device condition classification	1	1.0	1.0
UR18	Warranty information management	1	1.5	1.5
UR19	Secure role-based access control	2	2.0	4.0
UR20	Historical record retention	1	1.0	1.0
UR21	Concurrent appointment support	1	1.5	1.5
UR22	Efficient task completion	2	1	2.0

Project Totals

Item	Value
Total Project Effort (Person-Weeks)	72.0 Person-Weeks
Schedule (Effort Plus Buffer)	18.0 Weeks (72.0 pw ÷ 5 developers = 14.4 weeks raw + 25% buffer)
Weekly Salary Assumption	\$800 USD per developer per week
Cost	\$57,600 USD
Price (Cost Plus Margin)	\$69,120 – \$77,760 USD (20–35% margin)

Assumptions and Notes

Schedule Buffer (25%)

The raw schedule was calculated by dividing the total project effort (72.0 person-weeks) by the assumed team size of five concurrent developers, resulting in 14.4 weeks. A 25% schedule buffer was added, bringing the estimated project duration to 18.0 weeks.

The buffer accounts for integration overhead, requirement clarification, testing cycles, code reviews, and dependencies between requirements. For example, UR10 (user authentication) must be completed before protected functions such as UR2 (device purchasing), UR3 (device sale or exchange), UR11 (payment processing), and UR13 (complaint management) can be fully tested.

The 25% figure follows the course example and provides a reasonable contingency for a first-release software system of this scope.

The method is transparent and easy to audit: total effort divided by headcount, a 25% buffer applied, reasoning shown. Anyone reviewing this can follow the arithmetic without having to reverse-engineer your assumptions.

Price Margin (20–35%)

The quoted price range of \$69,120–\$77,760 applies a 20% lower margin and a 35% upper margin to the estimated development cost of \$57,600.

The lower margin represents a straightforward implementation with minimal requirement changes. The upper margin reflects uncertainty in more complex requirements, particularly UR2 (device purchasing and payment processing), UR11 (payment processing), and UR19 (role-based access control and security), which may require additional implementation effort and testing.

Using a price range rather than a fixed figure provides flexibility during project negotiation and reflects the uncertainty inherent in early-stage estimation.

Weekly Salary Assumption (\$800 USD per Developer per Week)

This figure follows the course example and is used consistently across all rows to allow comparison with other projects. It **represents a blended development rate for a junior-to-mid-level software developer working in a regional context.**

Rate justification, confidence band, and exclusion list are all present and consistent with each other. This is the most carefully argued section in the document - the quality here is noticeably higher than the UR writing.

In a real project, labour costs would vary depending on developer experience, employment arrangements, geographic location, and market conditions.

What Is Excluded From the Estimate

The estimate covers software development labour only.

The following items are explicitly excluded:

- Server and hosting infrastructure.
- Third-party software licences.
- External payment gateway fees related to UR2 and UR11.
- SMS or email notification service fees.
- Device inspection hardware or testing equipment.
- Staff training on the delivered system.
- Post-launch support and maintenance.
- Data migration from existing business records.

If any of these items are required, they would be scoped and priced separately.

Confidence in Per-UR Estimates

High-confidence estimates ($\pm 20\%$) include:

- UR6 (inspection information).
- UR17 (device classification).
- UR18 (warranty information).
- UR20 (historical record retention).
- UR21 (concurrent appointment support).

These requirements have relatively clear scope and limited workflow complexity.

Medium-confidence estimates ($\pm 40\%$) include:

- UR5 (device inspection and assessment).
- UR7 (appointment management).
- UR8 (repair request management).
- UR9 (delivery management).
- UR15 (business reporting).

These involve multiple actors and moderate workflow complexity.

Lower-confidence estimates ($\pm 60\%$) include:

- UR2 (device purchasing and payment processing).
- UR11 (payment processing).
- UR19 (security and role-based access control).

These requirements involve financial transactions, security controls, validation logic, and integration concerns that may introduce additional complexity during implementation.

Dominant URs

If effort on any single UR doubled, the requirements with the greatest impact on project cost and schedule would be:

- UR2 (purchase devices and complete payment processing) – 6.0 person-weeks.
- UR7 (appointment management) – 6.0 person-weeks.
- UR8 (repair request management and status tracking) – 6.0 person-weeks.
- UR11 (payment processing) – 6.0 person-weeks.

The team recommends that these requirements receive additional review during SR refinement so that hidden complexity is identified early.

Team Size Assumption

The schedule calculation assumes a team of five developers working in parallel whenever possible. Not all requirements can be parallelised because some have sequential dependencies.

For example:

- UR4 supports UR5.
- UR5 supports UR6.
- UR10 supports UR2, UR3, UR11, and UR13.

The estimate does not explicitly account for productivity loss caused by communication and coordination overhead among team members. This overhead is assumed to be partially absorbed by the 25% schedule buffer.